

Class Imbalance Mitigation for TCGA-UT

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July 14, 2026

Abstract

Class imbalance in computational pathology depends on the input regime: labeled image patches support ordinary classification, whereas whole-slide image (WSI) diagnosis is commonly modeled as weakly supervised learning over patch-feature bags. This report compares representative class-imbalance mitigation categories on TCGA-UT, a moderately imbalanced (head-to-tail $\approx 28:1$) histopathology dataset representative of real-world clinical class imbalance, using frozen Virchow2 representations for patch classification and WSI-bag classification on shared case-disjoint splits and three seeds. Each method is tuned over one imbalance-specific control selected on validation; held-out test results then rank the methods. Center-focused affinity loss (CFAL) reaches the highest patch-level mean macro F1; Odd-K-Out (OKO) set learning reaches the highest balanced accuracy and the best raw NLL and Brier among patch methods, although no patch method is uniformly best on every calibration metric; and CE plus soft-F1 with balanced sampling also improves both discrimination metrics, but by narrower margins. ProGAN augmentation and focal loss remain near the CE baseline. On WSI bags, RankMix has the highest mean balanced accuracy and MDE-MIL the highest mean macro F1; before temperature scaling MDE-MIL also has the best probability scores (lowest NLL, Brier, and ECE), but after a shared post-hoc step no WSI method is uniformly best on all calibration metrics, and the WSI gaps are small relative to seed-to-seed variation. Two lessons follow. First, once a strong frozen pathology encoder is fixed, patch-level imbalance gains come from reshaping the decision head or objective—as in CFAL’s prototype affinity loss or OKO’s set-level logit aggregation—rather than from resampling, difficulty reweighting, or synthetic-image augmentation, which barely move the CE baseline. Second, the most effective mechanism differs between the patch and WSI-bag regimes, so imbalance mitigation should be selected and evaluated in its native input regime alongside classwise recall and calibration, not treated as a universally transferable recipe.

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1 Introduction

Class imbalance changes what a pathology classifier is rewarded for learning. When common tumor types dominate training data, high aggregate accuracy can coexist with unreliable rare-class recall, especially if evaluation emphasizes prevalence-sensitive metrics. Long-tailed learning and medical-image evaluation literature therefore motivate classwise recall, macro F1, balanced accuracy, and calibration-aware probability metrics as primary outcomes in imbalanced settings (Saini and Susan, 2023; Zhang et al., 2023; Mosquera et al., 2024).

Imbalance methods often assume a specific prediction unit. Image-level methods use labeled images, image embeddings, or directly generated image samples, whereas WSI-level methods use weak labels over bags of instances together with attention, instance ranking, or bag-level representation learning. This report therefore evaluates each method family in the input form it is designed to consume. The broader computational-pathology pipeline these regimes sit in—whole-slide imaging, patching into tiles, and weakly supervised multiple-instance learning over frozen foundation-model features—is introduced in the companion datasets report (Queisler, 2026b).

TCGA-UT provides the inputs needed for both parts of the benchmark. The dataset includes labeled histopathology patches cropped from pathologist-selected tumor regions and slide identities that can be grouped into WSI-level bags. Patch labels are class labels, not dense pixel masks or cell-level annotations. This report uses that supervision directly: patch-level methods are evaluated on controlled patch classification with frozen Virchow2 embeddings, while MIL methods are evaluated on slide-level feature bags built from the same foundation-model family.

The purpose is not to introduce a new mitigation method or to reproduce every cited method in all implementation detail. Instead, the report compares the effectiveness of major class-imbalance mitigation categories on a current pathology foundation-model representation for both controlled patches and WSI feature bags. It implements representative data-level, loss-level, hybrid, synthetic-generation, and MIL-specific interventions under fixed splits and frozen Virchow2 features, tunes one imbalance-specific control per method on validation, and reports discrimination, classwise recall, and calibration on the held-out test split. Method fidelity is defined by implementing each method’s defining imbalance mechanism while holding the backbone family, split, input budget, and evaluation code fixed within each benchmark. Full method definitions and metric formulas are given in the companion methods report (Queisler, 2026a).

2 Dataset

TCGA-UT is the pan-cancer H&E cohort described in full in the companion datasets report (Queisler, 2026b); this section states only the properties the benchmark relies on. It contains 1,608,060 tumor *patches* of 256×256 pixels, cropped from pathologist-selected uniform tumor regions across 8,736 diagnostic *whole-slide images* (WSIs), each labelled with one of 32 *cancer types* (Tomczak et al., 2015; Komura et al., 2022). The patch labels are slide- or cancer-type labels, not dense tissue masks or cell-level annotations.

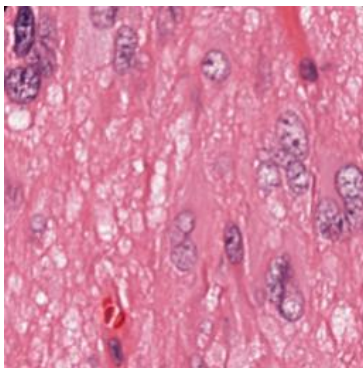
Imbalance is native to TCGA-UT rather than imposed after sampling: the largest class, glioblastoma multiforme (GBM; 792 slides), outnumbers the smallest, diffuse large B-cell lymphoma (DLBC; 28 slides), by a slide-level head-to-tail ratio of 28.3 (entropy-based imbalance $1 - H_{\text{norm}} \approx 0.09$; Queisler, 2026b). Spread over 32 classes this is a moderate, realistic long tail rather than an extreme one, which makes TCGA-UT representative of clinical cohorts and exposes the tension that motivates this study: methods must learn many visually related cancer categories while preserving performance on classes with far fewer slide-level examples.

Property	Value
Cancer-type labels	32
Diagnostic slides	8,736
TCGA-UT patches	1,608,060 patches of 256×256 pixels
Largest class	Glioblastoma multiforme (GBM), 792 slides
Smallest class	Diffuse large B-cell lymphoma (DLBC), 28 slides
Slide-level imbalance ratio	28.3
Shared split size per seed	5,025 train, 1,075 validation, 1,075 test cases; slide counts vary by seed
Leakage-control status	Case-disjoint by TCGA participant barcode; slide-disjoint follows from case assignment (Section 2.2)
Controlled patch input	Resolution level 0; up to 30 deterministic patches per slide; one frozen Virchow2 embedding per patch (Section 2.2)
WSI-bag input	Up to 30 deterministic frozen Virchow2 feature instances from the same slide identities

Table 1: Dataset properties used by the shared patch and WSI-bag benchmarks.

2.1 Data Units and Terminology

Several data units are involved in the benchmark, and the distinction matters for method comparison. A *slide* or *WSI* is one diagnostic histopathology image associated with one cancer-type label. A *case* or *patient* is the TCGA participant identified by the first three fields of a TCGA slide barcode, for example TCGA-OR-A5J1. A *tumor region* is a pathologist-selected area of a slide from which TCGA-UT crops groups of patches. A *patch* is one 256×256 RGB crop selected by the controlled manifest. A *patch embedding* is the frozen Virchow2 vector computed from that crop and consumed by the patch-level linear classifier. A *feature chunk* is a saved frozen-feature representation associated with a slide or slide subset. A *WSI bag* is the collection of frozen patch-feature instances used by the attention-MIL model for one slide-level prediction. *Head*, *body*, and *tail* classes refer to fixed support tiers derived from the full TCGA-UT slide distribution when reporting aggregate class-group metrics on the held-out test split. A *case-disjoint split* means that all slides, patches, and features from one TCGA participant stay in exactly one of train, validation, or test. Table 14 in Appendix B lists the same terms in compact form for quick reference.



Field	Example value
Cancer type	Glioblastoma multiforme
Resolution level	0
Patch size	256×256 RGB pixels
Source unit	One crop from a selected tumor region
Patch role	Image-level sample if selected by the controlled manifest
WSI-bag role	Candidate instance in the slide’s feature bag

Table 2: Example TCGA-UT data point copied from the shared raw dataset. The image is a single patch, while the table shows how that patch relates to the manifest fields used by the benchmark.

2.2 Shared Splits and Benchmark Inputs

For each seed, the split is assigned at the TCGA participant level rather than at the slide level. Case identifiers are derived from the first three TCGA barcode fields, so all diagnostic slides

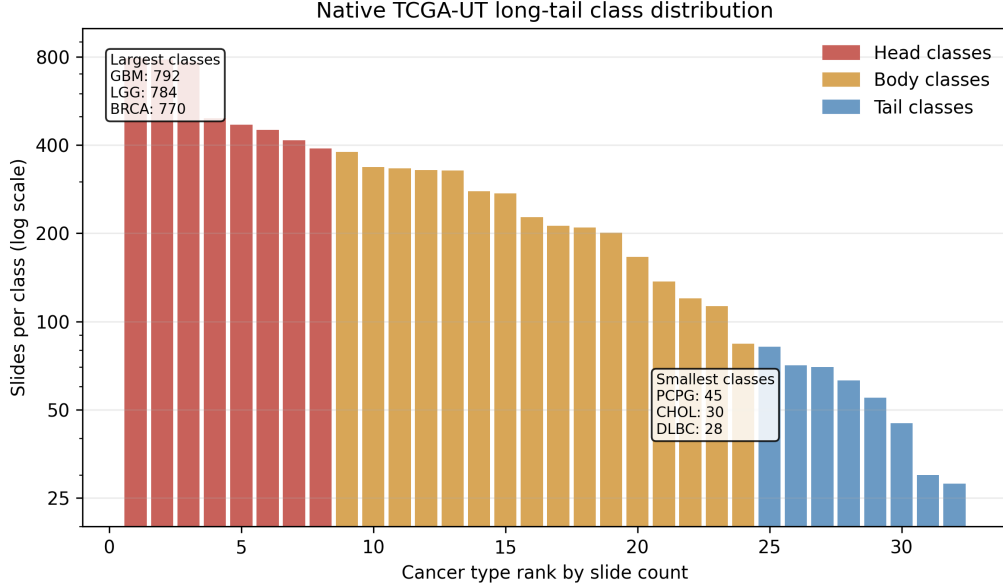


Figure 1: Native TCGA-UT slide support by cancer-type rank. The moderately imbalanced distribution motivates metrics that weight classes more evenly than aggregate accuracy.

from the same participant are kept in the same train, validation, or test partition. Every seed uses the same case-count budget—5,025 training, 1,075 validation, and 1,075 test cases—while the specific participants assigned to each partition differ by seed. Because cases contribute different numbers of diagnostic slides, the resulting slide counts vary across seeds: training contains 6,091–6,129 slides, validation 1,276–1,337 slides, and test 1,299–1,331 slides. The same case-level assignments are reused by both benchmarks, so no patches, frozen features, or WSI bags from the same slide or participant cross train, validation, and test.

This case-disjoint partition is the implemented leakage-control mechanism. It is stricter than slide-disjoint splitting alone, which would prevent patches from the same slide from crossing splits but would still allow different diagnostic slides from the same TCGA participant to appear in different partitions. The generated split reports record `case_id` as the split unit for all three seeds and no cross-split participant leakage after assignment. The patch benchmark and the WSI-bag benchmark see different input units, but they inherit the same train/validation/test case partition.

The patch benchmark derives a deterministic controlled patch manifest from the shared slide split. It uses resolution level 0 and samples up to 30 patches per slide with a fixed rule before method-specific training begins. The cap is a compromise between data coverage and slide-level control: it uses the empirical median number of raw JPG patches per slide at resolution level 0 (median 30, mean 31.1, range 10–80) while preventing slides with many extracted tumor-region patches from dominating the patch-level loss. At this resolution, 8,686 of 8,736 slides have at least 30 raw patches, so the cap retains the typical per-slide patch evidence while limiting only the larger slide folders. Each selected patch is encoded once with a frozen Virchow2 model, and all patch methods train the same lightweight classifier head on these embeddings (Zimmermann et al., 2024; Vorontsov et al., 2024). The encoder output is concatenated as `concat(CLS, mean(patch tokens))`, yielding 2,560-dimensional float16 vectors that are cast to float32 before classifier training. This design isolates imbalance mechanisms from both patch-selection policy and end-to-end image-backbone learning.

The WSI-bag benchmark applies the same per-slide evidence budget: when a frozen feature bag contains more than 30 instances, it is deterministically subsampled to 30 evenly spaced instances before attention-MIL training and evaluation. This keeps compute bounded and avoids

a systematic advantage for slides with larger extracted feature bags. The two benchmarks still differ in supervision unit and aggregation: patch models classify each selected crop independently, whereas WSI-bag models predict one slide label from an attention-weighted set of up to 30 instances.

3 Methods and Experimental Setup

The class-imbalance mitigation methods compared here are defined in full—mechanism, objective, and imbalance control—in the companion methods report (Queisler, 2026a). This section states only the concrete configuration used to run them on TCGA-UT: the shared classifier heads, the training budget, and the per-method validation grids. Cross-entropy (CE) is the no-mitigation reference in both regimes.

The comparison is regime-specific. Patch methods consume frozen Virchow2 embeddings of controlled patches and optimize one shared MLP classifier head; CE-based patch methods use the same linear output layer, while CFAL replaces it with class prototypes and affinity-based prediction. WSI-bag methods consume capped frozen Virchow2 feature bags and optimize one shared attention-MIL head. Both regimes rely on the same pathology foundation-model family and differ only in supervision unit, so method differences within a regime correspond to imbalance mechanisms rather than to changes in split, evaluation code, or encoder choice (Queisler, 2026a).

The tested methods, grouped by the class-imbalance taxonomy (Saini and Susan, 2023; Zhang et al., 2023), are:

- **Data-level sampling and augmentation:** balanced (inverse-frequency) sampling in both regimes, and RankMix teacher–student feature-subsequence mixing (WSI).
- **Algorithm-level losses:** class-weighted CE and focal loss in both regimes, and center-focused affinity loss (CFAL, patch).
- **Hybrid sampling–loss:** Scholz-style CE + soft-F1 and CE + soft-MCC (patch), Odd-K-Out (OKO) set learning (patch), and MDE-MIL dual-expert consistency (WSI).
- **Synthetic data generation:** per-class ProGAN patch augmentation (patch).
- **Representation and architecture:** SC-MIL supervised-contrastive bag learning (WSI).

Each non-baseline method varies one imbalance-specific control on validation (Table 4); the reference defaults and the selected configurations are reported in Section 5.1. Full method definitions and equations are given in the companion methods report (Queisler, 2026a).

3.1 Training Protocol and Reproducibility

Each method is evaluated under one shared training budget per benchmark (Table 3). Cross-entropy baselines use cited defaults without an imbalance-specific sweep. All other methods vary one imbalance-specific control on validation; the configuration with the highest mean validation macro F1 over seeds 0–2 is selected, with balanced accuracy as a tie-breaker. The test split never informs selection. Table 4 lists the reference default and validation grid for each control; the selected configurations and their tuning sensitivity are reported with the results (Section 5.1). ProGAN sweeps the number of final-depth (256×256) refinement epochs (Table 5); one long run produces snapshots at epochs 10, 25, and 50 so all sweep points share a single GAN training cost. The synthetic fraction is fixed at 1 (balance each tail class to the head-class count).

Figure 2 shows the patch and WSI-bag classifier heads.

The patch head is a two-layer MLP: linear map to 512 units, ReLU, dropout rate 0.1, and a linear output to 32 classes for CE-based methods. CFAL keeps the same trunk but replaces the

Setting	Patch benchmark	WSI-bag benchmark
Optimizer	AdamW	AdamW
Learning rate	10^{-3}	10^{-3}
Weight decay	10^{-4}	10^{-4}
Epochs	30	30
Early stopping	none	none
Minibatch size	256 patches	32 slide bags
Model selection	final epoch	final epoch
Random seeds	0, 1, 2	0, 1, 2

Table 3: Shared training protocol for both benchmarks. All methods within a benchmark use the same optimizer, schedule, and epoch budget unless noted in a method section.

Regime	Method	Reference default	Validation grid
Patch	Weighted CE	Weight power = 1	{0.25, 0.5, 0.75, 1}
WSI bag	Weighted CE	Weight power = 1	{0, 0.125, 0.25, 0.5, 0.75, 1}
Patch, WSI	Balanced sampler	Sampler power = 1	{0.5, 0.75, 1}
Patch, WSI	Focal loss	$\gamma = 2$	$\gamma \in \{0.5, 1, 1.5, 2\}$
Patch	CE + soft F1	Metric-loss weight = 1	{0.25, 0.5, 1, 2, 4, 8, 16, 32, 64, 256, 512, 1024}
Patch	CE + soft MCC	Metric-loss weight = 1	{0.25, 0.5, 1, 2, 8, 32}
Patch	CFAL	$\gamma = 2, \sigma = 1$	$\sigma \in \{0.25, 1, 4\}$
Patch	ProGAN augmentation	Final-depth epochs = 25	{10, 25, 50}
Patch	OKO	$k = 1$	$k \in \{1, 2, 3, 4, 5, 6, 8, 12, 20\}$
WSI bag	RankMix	$\alpha = 1$	$\alpha \in \{0.2, 0.5, 1, 2, 4, 8, 16, 32, 128\}$
WSI bag	SC-MIL	$\tau = 1$	$\tau \in \{0.05, 0.1, 0.2, 0.5\}$
WSI bag	MDE-MIL	$\lambda_{\text{con}} = 0.25$	$\lambda_{\text{con}} \in \{0.1, 0.25, 0.3, 0.5, 1, 2, 4\}$

Table 4: Method-specific imbalance controls. Each row varies one control on validation while retaining the shared budget in Table 3. ProGAN sweeps only the final-depth refinement epochs; generator training follows Table 5.

final linear map with one learnable prototype per class and Gaussian affinity-based prediction (Queisler, 2026a). The WSI-bag head is an attention-based MIL network on the same feature dimension: each instance passes through the shared encoder, instance weights come from a linear scorer followed by a softmax over instances, the bag representation is their weighted sum, and slide logits are produced by a final linear map. SC-MIL additionally routes bag embeddings through a two-layer projection head of width 256 with ReLU. MDE-MIL keeps the shared encoder and attention pooling but uses two separate expert heads on the bag embedding. The WSI-bag input is capped at 30 instances per slide using deterministic evenly spaced subsampling.

RankMix uses balanced slide-bag sampling during student training. Mixed bags are formed within each minibatch by pairing bags with a random permutation and mixing their teacher-ranked representative subsequences with a mixing coefficient $\lambda \sim \text{Beta}(\alpha, \alpha)$. The student is reinitialized after teacher preparation and trained for the same 30-epoch budget as the other MIL methods. MDE-MIL instead zips one natural-order batch with one balanced batch at every step, trains both experts on the shared aggregator without a teacher, and evaluates with mean-logit ensembling.

Head, body, and tail metrics group classes by support in the full TCGA-UT slide distribution rather than by held-out test counts. Cancer types are sorted by dataset slide count; the eight lowest-support classes form the tail tier, the eight highest-support classes form the head tier, and the remaining sixteen classes form the body tier. Tier-wise precision, recall, and F1 on validation and test splits then aggregate only over the predefined classes in each tier.

Setting	Value
Resolution stages	4×4 to 256×256 (seven stages)
Epochs per stage (depths 1–6)	25
Final-depth (256×256) refinement epochs	25 (reference); validation selects from $\{10, 25, 50\}$
Fade-in fraction for new layers	0.5 (pinned to the reference 25-epoch schedule)
Latent dimension	128
Base channel width	512
Adversarial loss	WGAN-GP ($\lambda = 10$, drift = 10^{-3})
Equalized learning rate	Yes (weight $\sim \mathcal{N}(0, 1)$, scale $\sqrt{2/\text{fan_in}}$ at forward)
Optimizer	Adam, learning rate 2×10^{-4} , $\beta_1 = 0.5$
Batch schedule	Depth-dependent schedule of Ruiz-Casado et al.
Real patches per class for GAN training	Up to 2,048, deterministic per-class subsample
Synthetic count per tail class	Head-class train count minus class train count

Table 5: ProGAN generator training settings. Depths 1–6 use 25 epochs each. For the final depth, one long run trains to 50 epochs and the generator is snapshotted at epochs 10, 25, and 50; validation selects the snapshot with the best mean macro F1 over seeds. This reuses the single GAN training run for all three sweep points at roughly $2\times$ the previous final-depth cost. Synthetic fraction is fixed at 1 (every tail class is balanced to the head-class count).

4 Evaluation

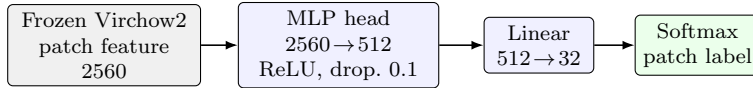
Both benchmarks use seeds 0, 1, and 2. Macro F1 is the primary ranking metric within each benchmark because it weights classes equally and penalizes failure on rare classes. Balanced accuracy, accuracy, classwise precision/recall/F1, and head/body/tail metrics are reported in the main result tables using the tier definitions in Section 3.1. These discrimination metrics answer whether the predicted class is usually correct, but they do not by themselves say whether the reported class probabilities are trustworthy, which matters in imbalanced medical classification where thresholding may depend on confidence rather than only on the argmax label (Mosquera et al., 2024). Three complementary probability-quality metrics—negative log likelihood, multiclass Brier score, and expected calibration error—are therefore reported on held-out softmax outputs, both before and after a unified post-hoc temperature-scaling protocol. Their definitions and the temperature-scaling protocol are given in the evaluation-framework section of the companion methods report (Queisler, 2026a). Patch and WSI-bag methods are never ranked against each other, because they answer related but different questions with different input units and supervision.

All reported results use validation-selected configurations evaluated on the held-out test split and show mean $\pm$ standard deviation over seeds. Tables 7 and 9 are the main comparison tables; Table 6 records the selected controls. Table 12 reports paired seed-wise differences for headline comparisons against each regime’s CE baseline; bracketed values list deltas in seed order 0–2 without formal significance testing or bootstrap confidence intervals. Cross-method gaps should therefore be interpreted from the held-out means together with the paired seed-wise directions. Large gaps with the same paired direction in all seeds provide stronger evidence; WSI-bag differences with overlapping standard deviations remain descriptive even when one method has the highest mean.

4.1 Probability Quality and Calibration

The three probability-quality metrics reported below—negative log likelihood (NLL), multiclass Brier score, and expected calibration error (ECE)—and the shared post-hoc temperature-scaling protocol are defined in the evaluation-framework section of the companion methods report (Queisler, 2026a). Each is computed once from raw final-epoch probabilities and once after

(a) Patch benchmark



(b) WSI-bag benchmark

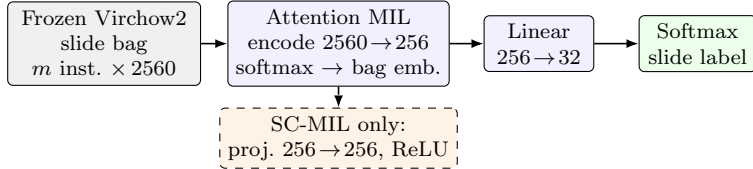


Figure 2: Shared classifier heads for the benchmark. Panel (a) classifies one controlled patch embedding per forward pass. CFAL reuses the MLP trunk but replaces the linear $512 \rightarrow 32$ head with learnable prototypes and affinity-based prediction. Panel (b) encodes all instances on a slide, pools them with attention, and predicts one slide-level label; the dashed branch is used only by SC-MIL. MDE-MIL reuses the same attention trunk but replaces the single linear head with two expert heads and mean-logit ensemble inference at evaluation. Virchow2 features are frozen in both benchmarks.

temperature scaling, where for each method and seed a single temperature is fit on validation by minimizing NLL and applied unchanged to that seed’s test scores. Temperature scaling leaves argmax rankings unchanged but corrects confidence scale, which is especially important for CFAL because its affinity outputs are ranking scores rather than likelihood-trained probabilities.

5 Results

Both benchmarks share case-disjoint splits, the Virchow2 feature family, and a maximum of 30 feature instances per slide, but they answer different prediction questions. The patch benchmark treats each selected crop as an independent labeled example; the WSI-bag benchmark predicts one slide label from an attention-weighted aggregation over the selected instances on that slide. Patch metrics therefore characterize patch-level classification only and must not be read as slide-level diagnostics; WSI-bag metrics characterize slide-level MIL only. Absolute scores and method orderings are not comparable across the two tables. Supporting diagnostic tables and figures referenced from this section are collected in Appendix A.

5.1 Selected Controls and Tuning Sensitivity

Each non-baseline method contributes one validation-selected imbalance control to the comparison. Table 6 records the control selected for each method, chosen by mean validation macro F1 over seeds 0–2 with balanced accuracy as a tie-breaker (Section 3.1); these configurations define the held-out test results reported in the remaining tables.

How much the selected control matters varies sharply across methods. A seed-blocked repeated-measures analysis of the validation sweeps shows that across-configuration variation exceeds twice the seed standard deviation only for OKO (range-to-seed-noise ratio 75.8) and RankMix (ratio 2.06, $p < 0.01$). For all single-knob resampling, reweighting, and focal mechanisms on frozen Virchow2 embeddings the swept control produces variation within seed noise regardless of its value; the selected configurations in Table 6 therefore reflect the best observed point rather than a robustly isolated tuning effect.

Regime	Method	Selected control	Val macro F1	Test macro F1	Test bal. acc.
Patch	CE	baseline (no sweep)	–	0.773	0.764
Patch	Balanced sampler	sampler_power=1	0.768	0.771	0.769
Patch	CE + soft F1 (balanced)	metric_loss_weight=256	0.781	0.780	0.781
Patch	CE + soft MCC (balanced)	metric_loss_weight=32	0.771	0.770	0.772
Patch	CFAL	cfal_sigma=1	0.792	0.798	0.789
Patch	Focal	focal_gamma=1.5	0.764	0.765	0.758
Patch	OKO	oko_k=4	0.783	0.788	0.791
Patch	ProGAN augmentation	final_depth_epochs=10	0.756	0.766	0.760
Patch	Weighted CE	weight_power=0.5	0.764	0.774	0.767
WSI bag	MIL CE	baseline (no sweep)	–	0.826	0.823
WSI bag	MDE-MIL (ensemble)	mde_mil_consistency_weight=2	0.851	0.865	0.849
WSI bag	Balanced MIL	sampler_power=0.75	0.821	0.831	0.829
WSI bag	Focal MIL	focal_gamma=0.5	0.823	0.824	0.823
WSI bag	Weighted MIL	weight_power=0.125	0.828	0.835	0.833
WSI bag	RankMix	rankmix_alpha=32	0.843	0.854	0.864
WSI bag	SC-MIL	sc_mil_temperature=0.1	0.828	0.839	0.837

Table 6: Validation-selected configuration provenance. Configurations are chosen by mean validation macro F1 over seeds 0–2; held-out test metrics in Tables 7 and 9 are reported after selection.

5.2 Patch-Level Results

Method	Accuracy	Balanced accuracy	Macro F1
CFAL	0.847 ± 0.008	0.789 ± 0.007	0.798 ± 0.009
OKO	0.841 ± 0.006	0.791 ± 0.005	0.788 ± 0.003
CE + soft F1 (balanced)	0.835 ± 0.004	0.781 ± 0.002	0.780 ± 0.003
Weighted CE	0.826 ± 0.008	0.767 ± 0.006	0.774 ± 0.005
CE	0.828 ± 0.010	0.764 ± 0.011	0.773 ± 0.015
Balanced sampler	0.827 ± 0.010	0.769 ± 0.007	0.771 ± 0.014
CE + soft MCC (balanced)	0.824 ± 0.009	0.772 ± 0.004	0.770 ± 0.005
ProGAN augmentation	0.824 ± 0.004	0.760 ± 0.005	0.766 ± 0.002
Focal	0.825 ± 0.007	0.758 ± 0.007	0.765 ± 0.009

Table 7: Validation-selected patch-level test results over three seeds. Methods share the same controlled manifest, frozen Virchow2 encoder, shared MLP trunk, and evaluation code.

The patch benchmark compresses into a narrow performance band once patches are represented with frozen Virchow2 embeddings (Table 7). CFAL ranks highest on macro F1, while OKO ranks highest on balanced accuracy. CFAL has paired gains over plain CE on both headline metrics in all three seeds (Table 12). OKO edges CFAL on balanced accuracy and stays above CE plus soft F1 on that metric, while achieving a macro F1 between CE plus soft F1 and CE plus soft MCC. CE plus soft F1 and OKO both improve on CE in the paired comparison, but the absolute gaps among these methods remain modest and standard deviations overlap in the summary table. Weighted CE, balanced sampling, and focal loss remain within the broader CE-centered band rather than separating cleanly from it.

The simpler patch interventions separate mainly by how little they move the validation-selected classifier after freezing Virchow2. Class-weighted CE, balanced sampling, focal loss, and ProGAN augmentation remain near CE on mean macro F1. ProGAN augmentation stays within the non-generative spread despite adding a large synthetic tail-class volume (Table 8).

One plausible explanation is the frozen-feature pipeline rather than poor generator quality. The class-specific generators reach an Inception FID of 239.5–382.0 (median 293.4), which is comparable to the ProGAN FID of 222–261 reported by Ruiz-Casado et al. on a binary breast-histology task where GAN augmentation still produced large downstream gains (Ruiz-Casado et al., 2024). The decisive difference is how the synthetic patches enter the classifier: the source method trains a convolutional classifier end to end on pixels, so it can adapt its features to the generated distribution, whereas this benchmark embeds every patch with a frozen Virchow2 encoder and trains only a lightweight head. A frozen encoder may map synthetic patches that are slightly off the real distribution into embedding regions the head rarely sees for real data, which would limit the value of additional synthetic volume even when pixel-level FID is reasonable. The per-class FID and Virchow2 nearest-neighbor distances in Table 8 are therefore reported as descriptive quality indicators rather than as proof of the null mechanism; the nearest-neighbor distances in particular lack a real-to-real baseline and should not be read as direct evidence of off-manifold generation. This frozen-feature explanation remains a hypothesis unless it is supported by feature-space real-versus-synthetic diagnostics such as real-to-real nearest-neighbor baselines, maximum mean discrepancy, or kNN class-mixing analysis. The small focal-loss shift is consistent with the interpretation that difficulty reweighting is not very helpful when most cancer types are already separable in the frozen embedding space.

Class	Real train	Generated	Inception FID	Virchow2 mean NN
Lymphoid Neoplasm Diffuse Large B-cell Lymphoma	600	16,240	345.6	50.68
Cholangiocarcinoma	660	16,180	307.6	51.79
Pheochromocytoma and Paraganglioma	900	15,940	382.0	54.06
Uveal Melanoma	1,160	15,680	338.4	48.88
Rectum adenocarcinoma	1,350	15,490	239.5	47.97
Mesothelioma	1,430	15,410	286.3	51.23
Uterine Carcinosarcoma	1,550	15,290	304.0	50.94
Kidney Chromophobe	1,740	15,100	282.8	48.97
All augmented classes (31 total)	165,470	356,570	293.4 median	51.32 median

Table 8: ProGAN augmentation diagnostics for seed 0. “Generated” counts are synthetic patches added per class to reach the head-class training patch count. Inception FID compares generated patches to the real training subset used for GAN training. Virchow2 mean NN is the mean, over a fixed synthetic subsample, of the minimum L_2 distance to real same-class embeddings in the frozen feature cache.

The classwise-recall plot (Figure 4) shows that observed errors are distributed across many cancer types rather than concentrated in a single tail class. In this setting, the main practical distinction is therefore not whether imbalance mitigation works at all, but which mechanism best preserves rare-class recall when the feature extractor is already strong. These conclusions are confined to the patch benchmark: high patch accuracy or macro F1 does not directly establish slide-level utility, because each test prediction is made on a single crop rather than by aggregating evidence across a slide bag.

5.3 WSI-Bag Results

The WSI-bag benchmark evaluates one slide-level prediction per diagnostic slide by aggregating capped feature bags from the shared split. The 30-instance cap aligns the per-slide evidence budget with the controlled patch manifest and prevents larger slides from benefiting simply by contributing more instances. The benchmark difference is architectural and supervisory: MIL

Method	Accuracy	Balanced accuracy	Macro F1
MDE-MIL (ensemble)	0.896 ± 0.002	0.849 ± 0.014	0.865 ± 0.011
RankMix	0.890 ± 0.005	0.864 ± 0.002	0.854 ± 0.008
SC-MIL	0.883 ± 0.013	0.837 ± 0.013	0.839 ± 0.008
Weighted MIL	0.882 ± 0.006	0.833 ± 0.019	0.835 ± 0.011
Balanced MIL	0.875 ± 0.003	0.829 ± 0.010	0.831 ± 0.009
MIL CE	0.873 ± 0.013	0.823 ± 0.014	0.826 ± 0.013
Focal MIL	0.872 ± 0.006	0.823 ± 0.019	0.824 ± 0.008

Table 9: Validation-selected WSI-bag test results over three seeds. Methods share the same capped feature bags, attention-MIL backbone family, and evaluation code.

learns slide-level attention over a set of instances, whereas the patch classifier treats each crop as an independently labeled example. Table 9 should be read as a within-regime slide-level leaderboard, not as confirmation or refutation of the patch-level ranking.

RankMix is above plain MIL CE on both headline metrics in all three paired seeds (Table 12) and has the highest mean balanced accuracy in this benchmark, although its macro-F1 gain is more modest than its balanced-accuracy gain and seed-wise standard deviations overlap other methods (Table 9, Figure 5). MDE-MIL reaches the highest mean test macro F1, with positive paired macro-F1 deltas in all three seeds but a wider spread in absolute macro F1 than RankMix. On raw probability quality, MDE-MIL has the lowest mean negative log likelihood (NLL), Brier score, and expected calibration error (ECE) among WSI methods, with RankMix second; both clearly improve on plain MIL CE before temperature scaling.

Other WSI methods cluster closely below this pair. Balanced-sampling MIL and SC-MIL trail RankMix and MDE-MIL in macro F1. SC-MIL performs well on validation but does not exceed plain MIL CE on the test macro-F1 average. Weighted MIL CE and focal MIL do not consistently improve over plain MIL CE. Focal MIL is in particular unstable across seeds: it is competitive with the stronger WSI methods on seeds 0 and 2 (test macro F1 0.846 and 0.845) but collapses on seed 1 (0.811), the single lowest per-seed point in the benchmark (Figure 5). Its seed-averaged macro F1 (0.834) is therefore driven by one low-outlier run rather than a uniform deficit, and its seed-to-seed standard deviation (0.020) is the widest among the WSI methods. Activation diagnostics confirm that the advanced WSI objectives were active: RankMix generated about 1,100,000 mixed examples per seed, SC-MIL constructed about 1,060,000 positive pairs per seed, and MDE-MIL processed both natural and balanced branches every training step.

5.4 Paired Seed Comparisons

Table 12 makes the headline gains explicit at seed resolution rather than only through aggregate means. CFAL is above plain CE on macro F1 and balanced accuracy in every seed, with larger shifts than the Scholz-style hybrids. CE plus soft F1 is above plain CE on balanced accuracy in every seed and on macro F1 in two of the three seeds, while CE plus soft MCC is positive on balanced accuracy in every seed but has one slightly negative macro-F1 delta. OKO is above plain CE on both headline metrics in every seed, with balanced accuracy gains comparable to CFAL and macro F1 gains between CE plus soft F1 and CE plus soft MCC. On WSI bags, both RankMix and MDE-MIL are above plain MIL CE on macro F1 and balanced accuracy in every seed. The absolute shifts are modest and would require formal paired testing to treat as significant.

5.5 Probability Calibration

Table 13 applies the metrics in Section 4.1 to the validation-selected configurations and makes the discrimination–calibration tradeoff explicit for raw outputs. On patches, the method with the lowest ECE is not the method with the strongest macro F1, and the Scholz-style hybrid objectives improve class-balanced discrimination without uniformly improving calibration. CFAL improves discrimination by the largest margin among patch methods but has the highest raw ECE and the worst NLL/Brier scores in Table 13. CFAL’s reported confidences are systematically too high relative to empirical accuracy—a typical over-confidence pattern when geometric affinities are passed through $\log a_{ic}$ and softmax—so it should be read as a strong patch-level *discriminator*, not as a calibrated probabilistic classifier on its raw outputs. OKO is the notable exception to the usual discrimination–calibration tradeoff on likelihood-based scores: it achieves the best raw NLL and Brier among patch methods while also placing highest on balanced accuracy. Its raw ECE, however, is not lowest: focal loss attains the smallest raw ECE, so no patch method dominates all three raw calibration metrics. The implicit logit regularization from averaging per-example outputs across set members limits how peaked the aggregate logit vector can become, which appears to prevent the overconfidence that afflicts CFAL and many CE variants. ProGAN augmentation further weakens probability quality relative to the CE baseline. On WSI bags, MDE-MIL and RankMix both raise mean balanced accuracy and sharply reduce raw ECE relative to plain MIL CE; MDE-MIL reaches the lowest raw NLL, Brier, and ECE, with RankMix close behind, while the other MIL variants cluster near the MIL CE baseline on all three raw metrics.

Regime	Method	NLL	Brier	ECE
Patch	CFAL	0.532 ± 0.029	0.220 ± 0.011	0.019 ± 0.003
Patch	OKO	0.617 ± 0.032	0.236 ± 0.011	0.029 ± 0.006
Patch	CE + soft F1 (balanced)	0.780 ± 0.026	0.263 ± 0.008	0.061 ± 0.028
Patch	Weighted CE	0.654 ± 0.030	0.252 ± 0.012	0.038 ± 0.005
Patch	CE	0.649 ± 0.035	0.250 ± 0.014	0.041 ± 0.002
Patch	Balanced sampler	0.653 ± 0.034	0.252 ± 0.013	0.038 ± 0.008
Patch	CE + soft MCC (balanced)	0.768 ± 0.034	0.271 ± 0.013	0.076 ± 0.029
Patch	ProGAN augmentation	0.723 ± 0.017	0.263 ± 0.005	0.073 ± 0.007
Patch	Focal	0.618 ± 0.025	0.251 ± 0.009	0.006 ± 0.004
WSI bag	MDE-MIL (ensemble)	0.471 ± 0.009	0.163 ± 0.003	0.023 ± 0.002
WSI bag	RankMix	0.428 ± 0.012	0.166 ± 0.005	0.028 ± 0.002
WSI bag	SC-MIL	0.440 ± 0.045	0.173 ± 0.018	0.007 ± 0.004
WSI bag	Weighted MIL	0.455 ± 0.033	0.177 ± 0.011	0.013 ± 0.004
WSI bag	Balanced MIL	0.479 ± 0.031	0.185 ± 0.010	0.013 ± 0.003
WSI bag	MIL CE	0.473 ± 0.018	0.188 ± 0.015	0.017 ± 0.010
WSI bag	Focal MIL	0.475 ± 0.024	0.189 ± 0.008	0.012 ± 0.002

Table 10: Held-out test calibration metrics for the validation-selected configurations after method-agnostic post-hoc temperature scaling. For each method and seed, the temperature is fit only on that seed’s validation split and then applied to the test split. This table is the decision-facing probability comparison; Table 13 is the raw-score diagnostic.

Table 10 reports post-hoc NLL/Brier/ECE after the shared temperature-scaling protocol. On patches, temperature scaling largely removes the raw-score inversion between CFAL and CE. CFAL reaches the best mean NLL and Brier score, while focal loss reaches the lowest mean post-hoc ECE. This is consistent with treating affinity outputs as rank scores that need a monotone probability map rather than as intrinsically miscalibrated classifiers. On WSI bags, RankMix has the lowest mean post-hoc NLL and MDE-MIL the lowest mean post-hoc Brier, with the two methods close on both. After scaling, however, the simpler MIL baselines—SC-

MIL, focal MIL, and plain MIL CE—reach lower mean ECE than either RankMix or MDE-MIL, so the raw RankMix and MDE-MIL ECE advantage over MIL CE should be read partly as a confidence-scale mismatch rather than as proof that they remain uniformly better calibrated after a shared post-hoc step.

All WSI-bag calibration scores are computed from the same held-out test slides, hard slide labels, and raw softmax outputs of the final-epoch model. An audit over stored test-result files confirms that all seven WSI methods share identical test label sequences within each seed, that reported probabilities sum to one, and that stored NLL, Brier, and ECE values match recomputation from the saved probability matrix.

5.6 Classwise Difficulty and Support

The classwise recall analysis shows that errors are uneven across cancer types, but support tiers alone do not reveal whether rare classes are always the hard classes. A post-hoc difficulty analysis therefore uses the CE baseline in each regime as the reference error pattern and compares each class’s full-dataset slide support with its seed-averaged held-out recall. Difficulty is operationalized as the false-negative rate, $1 - \text{recall}$, so the analysis asks whether classes with lower slide support are also the classes with higher baseline error.

Regime	Class	Slides	Support rank	Difficulty rank	Recall	F1
Patch	Rectum adenocarcinoma	63	5	1	0.103	0.144
Patch	Cholangiocarcinoma	30	2	2	0.319	0.409
Patch	Esophageal carcinoma	113	10	3	0.362	0.420
Patch	Uterine Carcinosarcoma	71	7	4	0.539	0.519
Patch	Mesothelioma	70	6	5	0.556	0.628
WSI-bag	Rectum adenocarcinoma	63	5	1	0.338	0.409
WSI-bag	Esophageal carcinoma	113	10	2	0.523	0.564
WSI-bag	Cholangiocarcinoma	30	2	3	0.528	0.544
WSI-bag	Uterine Carcinosarcoma	71	7	4	0.625	0.611
WSI-bag	Stomach adenocarcinoma	274	18	5	0.693	0.735

Table 11: Hardest classes under each regime’s CE baseline, averaged over seeds 0–2 on the held-out test split. Support rank is computed from full TCGA-UT slide counts, with rank 1 indicating the lowest-support class. Difficulty rank orders classes by baseline error rate, $1 - \text{recall}$.

Table 11 and Figures 7–8 make the support–difficulty mismatch explicit. Rectum adenocarcinoma is the hardest class in both regimes despite being the fifth-lowest-support class rather than the rarest class. Conversely, the rarest class, diffuse large B-cell lymphoma, does not appear among the five hardest classes in either regime. The rank association between dataset slide support and CE-baseline recall is positive but modest: Spearman $\rho = 0.334$ for patches and $\rho = 0.301$ for WSI bags. These descriptive correlations support a constrained interpretation: class frequency contributes to difficulty, but it does not fully determine which class boundaries remain hard in the frozen Virchow2 feature space.

6 Discussion

The two benchmarks support different but compatible conclusions. In the patch benchmark, a strong frozen encoder compresses most methods into a narrow band, but CFAL stands out most by replacing the linear head with prototype-based affinity training. OKO provides consistent gains on balanced accuracy comparable to CFAL and on macro F1 comparable to the Scholz-style hybrids, and it combines near-best discrimination with the best raw NLL and Brier among patch methods, although focal loss has lower raw ECE. The implicit mechanism is logit

averaging across set members: each set prediction is the sum of individual forward passes, which limits how peaked the aggregated logit can become and thereby suppresses the overconfidence that affects CE-trained classifiers. The Scholz-style hybrid objectives provide modest improvements over plain CE rather than large separations, and the simpler reweighting or resampling variants remain within the broader CE overlap band. ProGAN augmentation does not deliver a clear advantage even though it adds large numbers of synthetic tail-class patches; one plausible explanation is the frozen-feature pipeline, since the generators reach an Inception FID comparable to the source study that reported large gains, and that study adapted its classifier to the synthetic images end to end whereas this benchmark consumes them through a frozen encoder and a lightweight head. That explanation remains a hypothesis unless feature-space real-versus-synthetic diagnostics confirm it. Focal loss is also unhelpful here, consistent with the interpretation that difficulty reweighting is redundant when the representation has already absorbed much of the class-separation problem. In the WSI-bag benchmark, RankMix has the highest mean balanced accuracy and is above MIL CE on balanced accuracy in every seed, while MDE-MIL reaches the highest mean macro F1 and the best raw probability scores, though with higher seed variance; these WSI gaps are small relative to the seed-to-seed spread.

These readings should be interpreted within the benchmark’s deliberate scope. The experiment is controlled rather than a full reproduction of every original paper: method sections document the necessary adaptations to frozen Virchow2 features and TCGA-UT supervision, and both regimes use frozen features rather than end-to-end encoder adaptation. Each non-baseline method is tuned over one imbalance-specific control on validation; the search is compact rather than an exhaustive optimization of every published implementation. The resulting rankings therefore describe how imbalance mechanisms behave when patch and slide classification are already supported by a strong pathology foundation model.

The dataset’s moderate imbalance also means that class frequency is only one part of the difficulty signal. The classwise difficulty analysis shows this directly: several low-support classes are hard, but the hardest classes are not exactly the rarest classes, and support explains only a modest share of the classwise recall ordering. This distinction helps explain why pure exposure corrections have limited room to improve the patch benchmark. Balanced sampling and class-weighted CE increase the influence of observed minority examples, but they do not add new within-class diversity or change the separability of visually related classes in the Virchow2 feature space.

Holding the encoder fixed within each benchmark therefore changes what an imbalance intervention can accomplish. On patches, the bottleneck shifts from learning stain and tissue morphology to calibrating decision boundaries under an imbalanced label distribution. Under that bottleneck, CFAL’s prototype shaping provides the largest discrimination gain, whereas objectives that reshape training at the set or minibatch level—OKO through set-aggregated logit regularization, and the Scholz-style hybrids through metric-derived loss terms—provide more moderate discrimination improvements; OKO additionally yields the best raw NLL and Brier among patch methods. On slides, the bottleneck instead includes instance selection and bag aggregation, so RankMix can add value by mixing informative subsequences even when the underlying features are strong. MDE-MIL shows that coupling natural and balanced branches with cross-expert consistency can raise mean macro F1 and gives the best raw probability scores, although the macro-F1 gain is seed-sensitive and RankMix has the highest mean balanced accuracy in this benchmark.

These results should be read within each benchmark’s prediction unit. Patch accuracy does not directly imply slide-level deployment quality, and WSI-bag gains may reflect aggregation behavior absent from the patch classifier. The shared split, encoder family, and 30-instance slide budget link the two benchmarks descriptively, but the evidence for each method comes from the table corresponding to its evaluated input.

Several limitations apply. TCGA-UT patch labels come from pathologist-selected tumor

regions; they are useful patch-level class labels but not dense annotations. The two benchmarks share case-disjoint splits but differ in prediction unit and per-slide evidence budget, so their result tables are non-transferable. The validation sweep varies one imbalance-specific control per method and does not exhaust architecture, training schedule, multiple interacting hyperparameters, or remaining ProGAN generator-side choices beyond the final-depth refinement epochs (latent dimension, base channels, and learning rate). In particular, ProGAN augmentation is evaluated only through the frozen Virchow2 pipeline; whether GAN-generated patches help under end-to-end encoder adaptation is left open and is the most plausible route to a non-null synthetic-augmentation result on this dataset.

7 Conclusion

TCGA-UT provides a native, moderately imbalanced setting representative of real-world clinical imbalance for comparing representative imbalance interventions on frozen Virchow2 patch and WSI-bag representations. After validation-selected tuning on case-disjoint splits, CFAL reaches the highest patch-level mean macro F1; OKO set learning reaches the highest balanced accuracy while achieving the best raw NLL and Brier among patch methods; and CE plus soft F1 with balanced sampling also improves both discrimination metrics, but with narrower gaps. No patch method is uniformly best across raw and post-hoc calibration metrics. ProGAN augmentation and focal loss remain close to the CE baseline. On WSI bags, RankMix has the highest mean balanced accuracy and MDE-MIL the highest mean macro F1, with MDE-MIL also giving the best raw probability scores (lowest NLL, Brier, and ECE); these WSI gaps are small relative to seed variation, so they should be read as the highest means in this benchmark rather than as established differences. The practical implication is that imbalance mitigation should be evaluated in the native input regime and alongside classwise recall and calibration, not only by aggregate discrimination.

A Additional Result Tables and Figures

ProGAN synthetic patches and nearest real training neighbors (seed 0, Virchow2 embedding space)

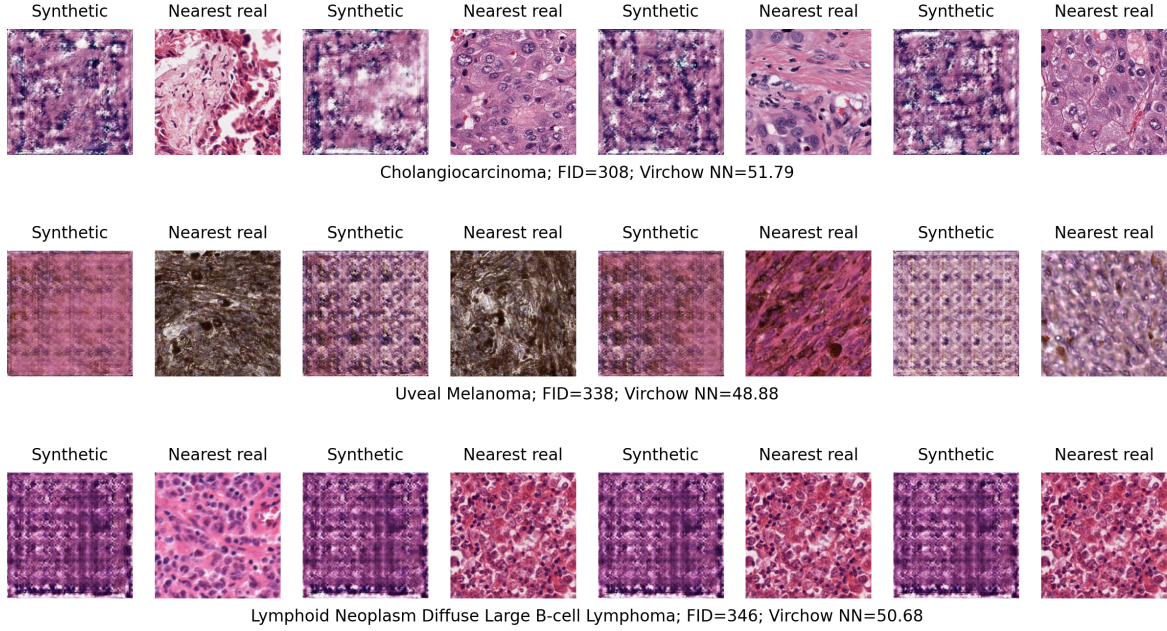


Figure 3: ProGAN synthetic patches (left column of each pair) and their nearest real training neighbors in Virchow2 embedding space (right column) for three tail classes in seed 0. Visual quality is class-dependent: the Cholangiocarcinoma generator still produces clear color-banding artifacts, whereas the Uveal Melanoma and Lymphoid Neoplasm generators yield patches that look closer to real histology. Visual quality is therefore not the limiting factor: the per-class Inception FID values are comparable to those reported for ProGAN in the source augmentation study, and ProGAN augmentation is interpreted through the frozen-feature pipeline—synthetic patches are embedded by a frozen Virchow2 encoder and consumed by a lightweight head—rather than through pixel-level appearance or FID alone.

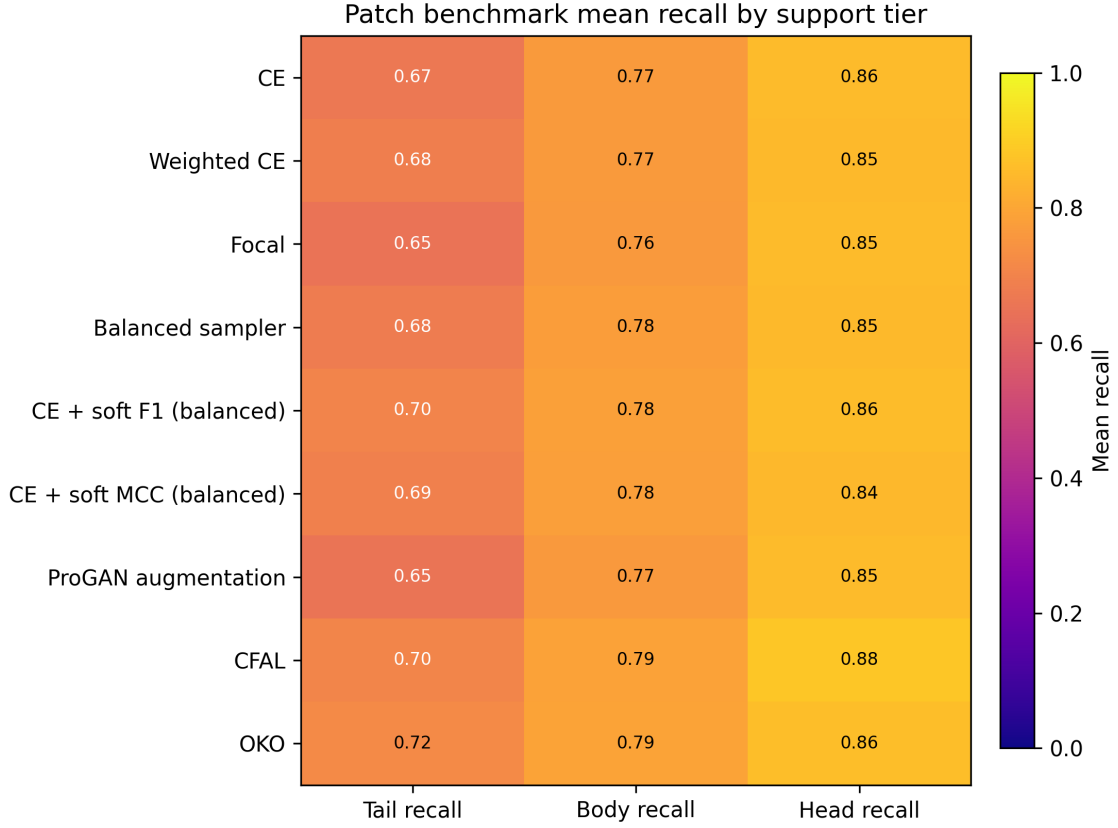


Figure 4: Patch-level mean recall by support tier on the held-out test split. The heatmap exposes the head–body–tail variation hidden by aggregate accuracy and motivates reporting balanced accuracy and macro F1.

Comparison	Δ Macro F1	Δ Balanced accuracy
CE + soft MCC – CE	-0.003 ± 0.008 [+0.005, −0.000, −0.014]	$+0.009 \pm 0.007$ [+0.015, +0.012, −0.000]
CE + soft F1 – CE	$+0.007 \pm 0.010$ [+0.014, +0.014, −0.007]	$+0.018 \pm 0.008$ [+0.023, +0.023, +0.007]
OKO – CE	$+0.015 \pm 0.010$ [+0.024, +0.021, +0.001]	$+0.028 \pm 0.012$ [+0.038, +0.033, +0.011]
CFAL – CE	$+0.025 \pm 0.005$ [+0.030, +0.027, +0.019]	$+0.026 \pm 0.004$ [+0.030, +0.027, +0.020]
RankMix – MIL CE	$+0.028 \pm 0.010$ [+0.023, +0.041, +0.020]	$+0.041 \pm 0.010$ [+0.038, +0.055, +0.031]
MDE-MIL – MIL CE	$+0.039 \pm 0.016$ [+0.055, +0.043, +0.018]	$+0.026 \pm 0.016$ [+0.041, +0.034, +0.003]

Table 12: Paired validation-selected test-split differences against the within-regime CE baseline on seeds 0–2. Each cell reports mean $\pm$ SD across seeds; bracketed values list seed-ordered paired deltas. No formal p -values are computed.

Regime	Method	NLL	Brier	ECE
Patch	CFAL	2.850 ± 0.011	0.914 ± 0.001	0.787 ± 0.008
Patch	OKO	0.756 ± 0.040	0.270 ± 0.015	0.158 ± 0.013
Patch	CE + soft F1 (balanced)	3.038 ± 0.137	0.316 ± 0.008	0.155 ± 0.004
Patch	Weighted CE	1.965 ± 0.115	0.314 ± 0.016	0.147 ± 0.007
Patch	CE	1.976 ± 0.149	0.312 ± 0.018	0.147 ± 0.009
Patch	Balanced sampler	1.967 ± 0.139	0.313 ± 0.019	0.147 ± 0.009
Patch	CE + soft MCC (balanced)	2.870 ± 0.153	0.332 ± 0.017	0.162 ± 0.008
Patch	ProGAN augmentation	2.511 ± 0.093	0.327 ± 0.007	0.157 ± 0.004
Patch	Focal	1.370 ± 0.081	0.298 ± 0.014	0.131 ± 0.006
WSI bag	MDE-MIL (ensemble)	0.487 ± 0.005	0.166 ± 0.002	0.036 ± 0.004
WSI bag	RankMix	0.584 ± 0.029	0.170 ± 0.006	0.049 ± 0.001
WSI bag	SC-MIL	1.024 ± 0.145	0.204 ± 0.020	0.092 ± 0.009
WSI bag	Weighted MIL	1.224 ± 0.144	0.212 ± 0.012	0.099 ± 0.006
WSI bag	Balanced MIL	1.264 ± 0.165	0.223 ± 0.010	0.102 ± 0.005
WSI bag	MIL CE	1.283 ± 0.080	0.227 ± 0.024	0.106 ± 0.012
WSI bag	Focal MIL	1.117 ± 0.153	0.221 ± 0.014	0.098 ± 0.009

Table 13: Held-out test calibration metrics for the validation-selected configurations over three seeds from raw final-epoch probabilities. Lower is better for negative log likelihood (NLL), multiclass Brier score, and expected calibration error (ECE). Values are not comparable across regimes because patch and WSI-bag models operate on different input units.

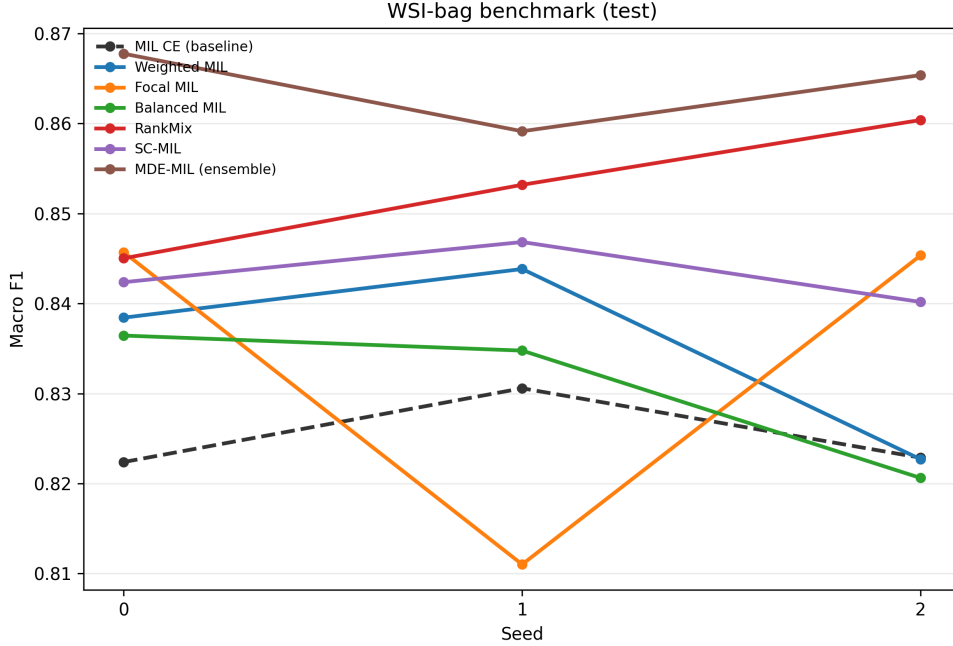


Figure 5: Validation-selected WSI-bag macro F1 on the held-out test split by random seed. The dashed line marks plain MIL CE; seed-wise spread shows why cross-method gaps should be read cautiously when standard deviations overlap (Table 9). The pronounced focal-MIL dip on seed 1 is a single-seed training instability: focal MIL tracks the strongest methods on seeds 0 and 2 but collapses on seed 1, so its averaged result reflects one unstable run rather than a consistent deficit.

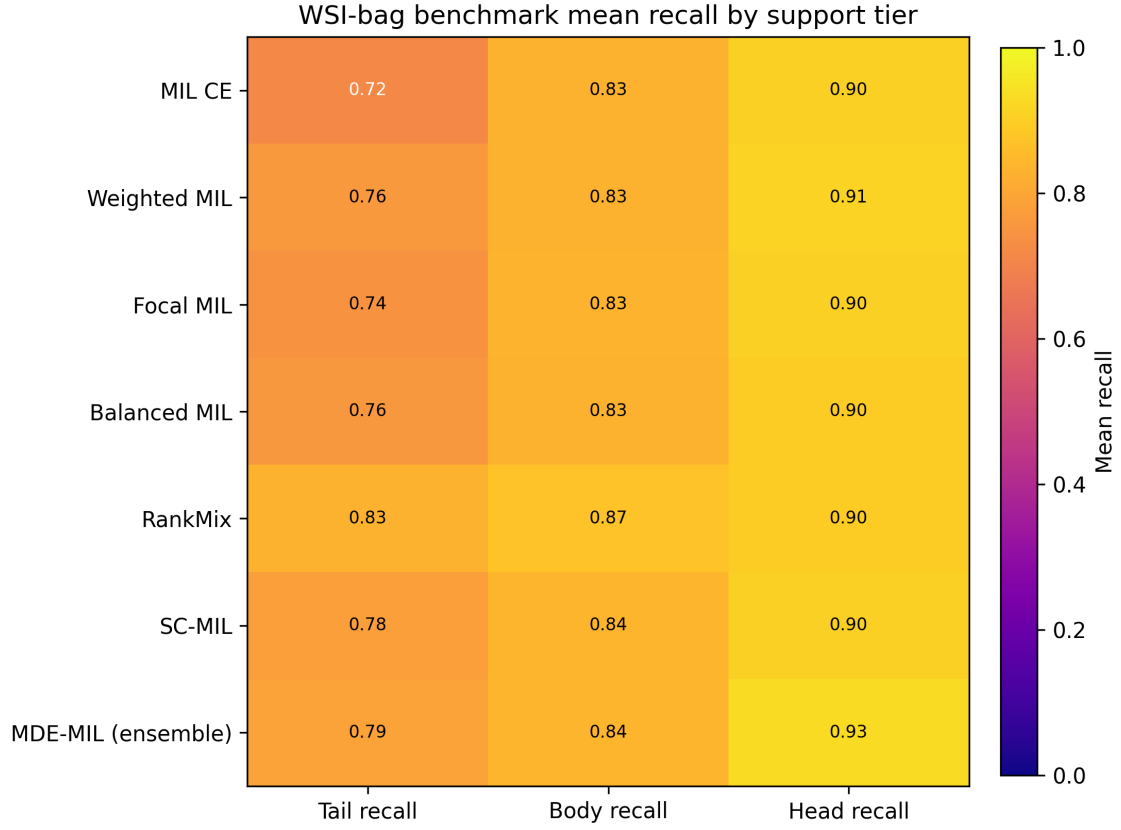


Figure 6: WSI-bag classwise recall on the held-out test split. The spread across classes shows why aggregate accuracy alone is insufficient for the imbalanced setting.

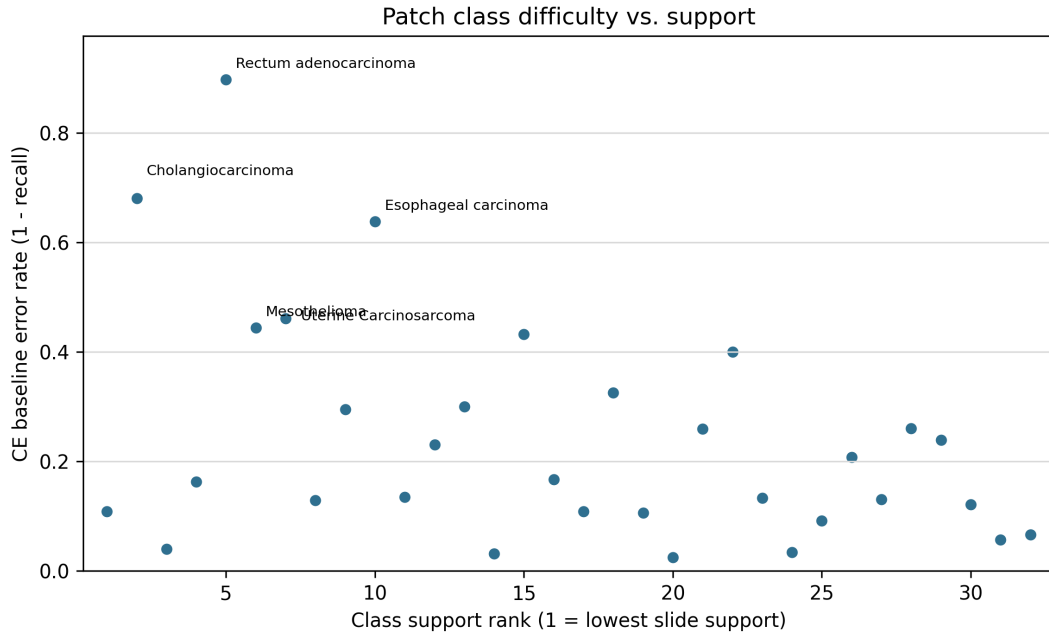


Figure 7: Patch CE-baseline class error rate versus full-dataset class-support rank. The hardest patch classes are not simply ordered by rarity, indicating that exposure-based corrections address only part of the error pattern.

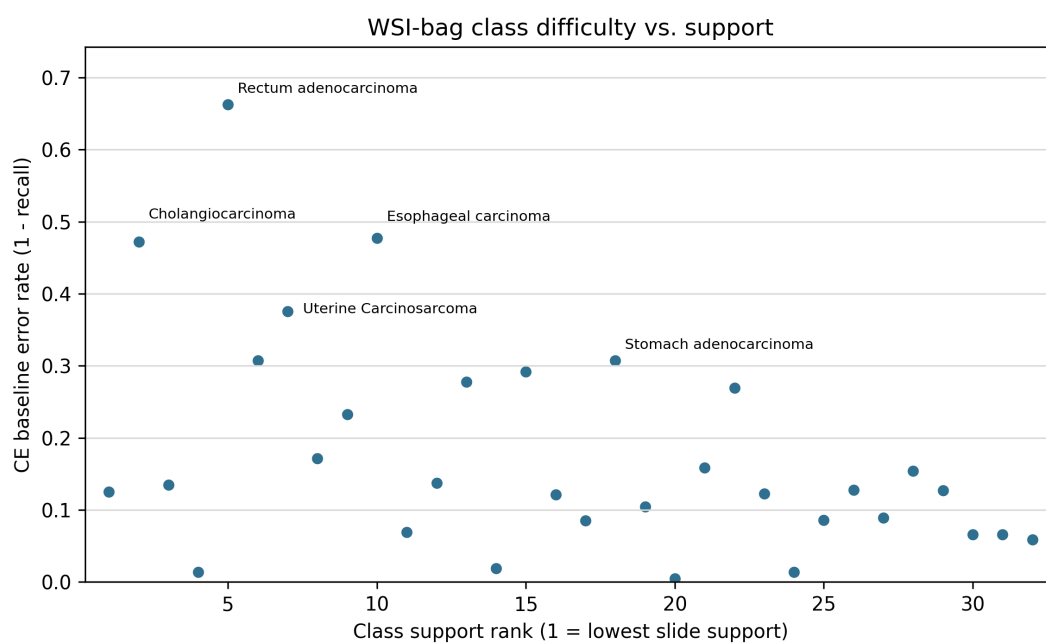


Figure 8: WSI-bag CE-baseline class error rate versus full-dataset class-support rank. The same high-error classes are visible at slide level, showing that class difficulty is not reducible to class frequency.

B Terminology

Term	Meaning in this study
<i>Slide / WSI</i>	Diagnostic whole-slide image carrying one cancer-type label.
<i>Case / patient</i>	TCGA participant identified by the first three fields of the TCGA slide barcode, such as TCGA-OR-A5J1.
<i>Patch</i>	256×256 H&E crop from a selected tumor region; patch labels inherit the slide cancer type.
<i>Patch embedding</i>	Frozen Virchow2 vector for one controlled patch; input to the patch-level classifier head.
<i>Region</i>	Tumor area selected before patch extraction.
<i>Cancer type</i>	The 32-way class label used for patch and slide-level classification.
<i>Feature chunk</i>	Frozen Virchow2 feature representation used to assemble slide-level feature bags.
<i>WSI bag</i>	Set of patch-feature instances for one slide, consumed by the MIL model.
<i>Head/body/tail class</i>	Fixed support tier from the full TCGA-UT slide distribution; see Section 3.1.
<i>Case-disjoint split</i>	Split rule preventing any TCGA participant’s slides, patches, or features from crossing train, validation, and test.

Table 14: Terminology used for the two native input regimes (see Section 2.1).

References

- Komura, D., Kawabe, A., Fukuta, K., Sano, K., Umezaki, T., Koda, H., Suzuki, R., Tominaga, K., Ochi, M., Konishi, H., et al. (2022). Universal encoding of pan-cancer histology by deep texture representations. *Cell Reports*, 38(9):110424.
- Mosquera, C., Ferrer, L., Milone, D. H., Luna, D., and Ferrante, E. (2024). Class imbalance on medical image classification: towards better evaluation practices for discrimination and calibration performance. *European Radiology*, 34(12):7895–7903.
- Queisler, Y. (2026a). Class imbalance mitigation: Methods and evaluation framework. Technical report, TU Berlin.
- Queisler, Y. (2026b). Computational pathology and the benchmark datasets. Technical report, TU Berlin.
- Ruiz-Casado, J. L., Molina-Cabello, M. A., and Luque-Baena, M. R. (2024). Enhancing histopathological image classification performance through synthetic data generation with generative adversarial networks. *Sensors*, 24(12):3777.
- Saini, M. and Susan, S. (2023). Tackling class imbalance in computer vision: a contemporary review. *Artificial Intelligence Review*, 56(S1):1279–1335.
- Tomczak, K., Czerwinska, P., and Wiznerowicz, M. (2015). The cancer genome atlas (tcga): an immeasurable source of knowledge. *Contemporary Oncology*, 19(1A):A68–A77.
- Vorontsov, E., Bozkurt, A., Casson, A., Shaikovski, G., Zelechowski, M., Severson, K., Zimmermann, E., et al. (2024). A foundation model for clinical-grade computational pathology and rare cancers detection. *Nature Medicine*, 30(10):2924–2935.
- Zhang, Y., Kang, B., Hooi, B., and Yan, S. (2023). Deep long-tailed learning: A survey. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 45(9):10795–10816.
- Zimmermann, E., Vorontsov, E., Viret, J., Casson, A., Zelechowski, M., and Shaikovski, G. (2024). Virchow2. *2024 Shell TechXplorer Digest*, 2024(1).